Predicting the 2026 FIFA World Cup: A Full Project Report Across Match Outcome Models

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Code repository: `https://github.com/ilirbejleri/world-cup-ml-final`

Abstract

This project builds an end-to-end machine learning system for FIFA World Cup prediction. The pipeline is self-contained in the `final/` directory [Ahmad et al., 2026] and converts cached historical football data into 14,881 chronological supervised match examples: 964 World Cup rows and 13,917 other international fixtures. Every feature is computed before kickoff. Every model is selected through walk-forward World Cup checkpoints: train on all matches available before a World Cup, validate on that next tournament, and repeat this for every feasible pre-2022 World Cup from 1938 through 2018. The 2022 World Cup is held out as the final test. We compare original MLR feature sets, expanded MLR/RF/NN variants, independent Poisson score models, draw-inflated Poisson models, and a validation-tuned ensemble. The best held-out log loss is achieved by an 18-feature neural network using all original engineered feature families (0.9321 on 2022). The best held-out accuracy is achieved by a 53-feature NN ablation that removes the two noisiest underdog-history features (60.9% accuracy, 0.9363 log loss, and 30% draw recall). A classifier-calibrated Poisson score simulator powers the 5,000-run 2026 tournament forecast: the best log-loss NN supplies W/D/L probabilities, and the score layer supplies calibrated goal counts. The project shows that supervised international matches are necessary for high-capacity models. For the 2026 forecast, the pipeline updates Elo, form, the outcome model, and the score layer with 4,358 supplemental matches from 2022 through March 2026; Spain has the highest simulated champion probability at 14.9%.

1 Introduction

The FIFA World Cup is an attractive machine learning problem because it is global, high-stakes, and structured, but it is also a difficult prediction target. A modern tournament has only 64 matches before 2026, and the expanded 2026 tournament still has far fewer observations than ordinary supervised ML settings. Football is also low scoring: a model can correctly identify the stronger team and still miss a draw, penalty shootout, or single-goal upset. For this project, accuracy and log loss both matter. Accuracy measures whether the model selects the right outcome; log loss measures whether its probability distribution is useful for simulation.

The central data-design choice is to treat international matches as supervised labels rather than only as rating-history updates. Every international match in the cache is a supervised example, while World Cup matches receive higher sample weights so the target remains tournament behavior. The report describes the complete system: data construction, engineered features, MLR/RF/NN/Poisson comparisons, validation checkpoints, error analysis, and a 2026 tournament forecast.

The project has four goals. First, compare multiple model families under one leakage-resistant chronological protocol. Second, show exactly how the 18 engineered feature families are represented and how many each model uses. Third, produce at least one strong high-accuracy model and

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one low-log-loss probabilistic model. Fourth, use a hybrid outcome/score simulator to forecast the configured 2026 World Cup field.

2 Background

Each match is represented from the perspective of the pre-match higher-Elo team. The labels are Win, Draw, and Loss for that stronger side, so a Loss is an upset. This framing avoids dependence on arbitrary home/away ordering in neutral-site tournaments. Log loss is

$$\mathrm{LL} = -\frac{1}{N}\sum_{i=1}^{N} \log \hat{p}_{i,y_i},$$

where $\hat{p}_{i,y_i}$ is the predicted probability assigned to the true class. Log loss is the primary calibration metric because tournament simulation uses probabilities rather than only argmax outcomes.

The model families reflect different assumptions. Multinomial logistic regression (MLR) is a stable softmax baseline. Random forests (RF) capture nonlinear feature interactions but can be less calibrated on small validation sets [Breiman, 2001]. Neural networks (NNs) can combine many rolling-window signals but require enough supervised examples. Poisson score models directly predict goals and derive win/draw/loss probabilities from a score grid, following classical football score modeling [Maher, 1982, Dixon and Coles, 1997]. Elo ratings are used because they have been shown to be useful in association-football prediction [Hvattum and Arntzen, 2010].

3 Setting

Data. The final directory includes the raw match cache, feature code, report source, style file, and bibliography. The supervised table contains 14,881 matches. The base cache ends with internationals through 2021 and the 2022 World Cup. Those rows are used for model comparison and 2022 testing. For the 2026 forecast only, we add a bundled supplemental international-results file covering 2022 through March 2026 [Jürisoo, 2026]. Table 1 summarizes the base splits.

Table 1: Final supervised dataset. International rows are training labels, not only rating-history updates.

Split	Rows	World Cup	International	Dates
All supervised rows	14,881	964	13,917	1930-07-13 to 2022-12-18
Pre-2022 training pool	14,817	900	13,917	1930-07-13 to 2021-12-31
Held-out 2022 World Cup	64	64	0	2022-11-20 to 2022-12-18

Walk-forward checkpoints. The validation protocol uses every feasible World Cup checkpoint before 2022. For a checkpoint year Y, models train on matches before the first match of World Cup Y and validate on World Cup Y. Checkpoints begin in 1938 because 1934 lacks all three classes in the prior training data. The final 2022 World Cup remains untouched until the end. This design directly answers the assignment requirement that training on a set of matches be tested on the next upcoming World Cup repeatedly.

Table 2: Checkpoint protocol. Full per-model rows are written to `outputs/fold_metrics.csv`.

Checkpoint range	WCs	First train rows	Last train rows
1938–2018 validation	19	35	12,551
2022 final test	1	14,817	14,817

4 Methodology

Chronological features and weights. A feature tracker processes matches in date order. For each row it records pre-match Elo, rolling form, goals scored, goals conceded, clean sheets, opponent

strength, head-to-head history, World Cup history, and match context before updating state with the result. World Cup rows receive base weight 5.0, other internationals receive base weight 0.75, and both receive a 12-year half-life decay:

$$w_i = w_{\text{competition}(i)} \cdot 0.5^{(Y-y_i)/12}.$$

This prevents old matches and lower-importance friendlies from overwhelming the tournament target.

Eighteen engineered feature families. The feature design is organized around 18 engineered families. Table 3 shows the families; the appendix contains the full glossary. A is the higher-Elo team before kickoff, and B is the lower-Elo team.

Table 3: Original engineered feature families used for model accounting.

Family	Features
Team strength	`elo_a`, `elo_b`, `wc_hist_a`, `wc_hist_b`
Recent form and scoring	`wr_a`, `wr_b`, `gd_a`, `gd_b`, `gs_a`, `gs_b`
Schedule and defense	`opp_elo_a`, `opp_elo_b`, `cs_a`, `cs_b`
Matchup and context	`h2h_a`, `stage_ord`, `upset_a`, `gk_b`

How 18 becomes 55. The 55-column full models are not 55 unrelated ideas. They expand the 18 families into multiple time windows and interaction terms. The strength block contains eight columns: raw Elo for A/B, Elo difference, Elo win probability, World Cup history for A/B, and recent opponent strength for A/B. Thus the column count is explicit: $8 + 12 + 12 + 12 + 11 = 55$. Recent form is expanded into three windows: last 5 matches, last 10 matches, and recent 4-year form. Each window includes win rate, goal differential, goals scored, goals conceded, clean-sheet rate, and differences between A and B. Context adds head-to-head features, stage indicators, World Cup indicators, absolute strength gaps, squared Elo gap, and two underdog-history exponential moving averages (EMAs): favorite-upset rate and underdog giant-killing rate. The exact mapping is saved in `outputs/feature_blocks_55.csv`.

Table 4: The 55 direct full-model columns are grouped into five blocks.

Block	Columns	Purpose
Strength/history	8	Long-run team quality and tournament history
Last-5 form	12	Short-term performance and opponent quality
Last-10 form	12	More stable medium-window form
Recent 4-year form	12	Cycle-aware national-team form
Context/interactions	11	Stage, H2H, nonlinear gaps, upset/underdog signals

Model variants. We retrain the original MLR feature sets under the new checkpoint protocol: baseline-5, original-18, form-only, and matchup-only. We also train expanded MLR, RF, and NN variants: strength, full, and recent-no-Elo. The NN family adds an original-18 model and a 53-feature no-underdog ablation that removes `upset_a` and `gk_b`. Score models include independent Poisson regression and a draw-inflated Poisson variant. The ensemble chooses one model per family based only on validation checkpoints and tunes convex weights on those checkpoints.

Knockout draw handling. Draws are valid in group matches, but not as final knockout outcomes. The pipeline explicitly handles this in two places. During validation and testing, if a row is a knockout stage, predicted draw probability is set to zero and win/loss probabilities are renormalized. During 2026 simulation, knockout score grids are calibrated after setting draw probability to zero, so sampled knockout scorelines are decisive; a stochastic tiebreak fallback remains for numerical edge cases. This includes the 2026 round of 32.

5 Experiments

Table 5 reports the final held-out 2022 results after retraining on all feasible World Cup checkpoints. The best log loss is NN-original-18 at 0.9321, which is useful because it shows the original feature

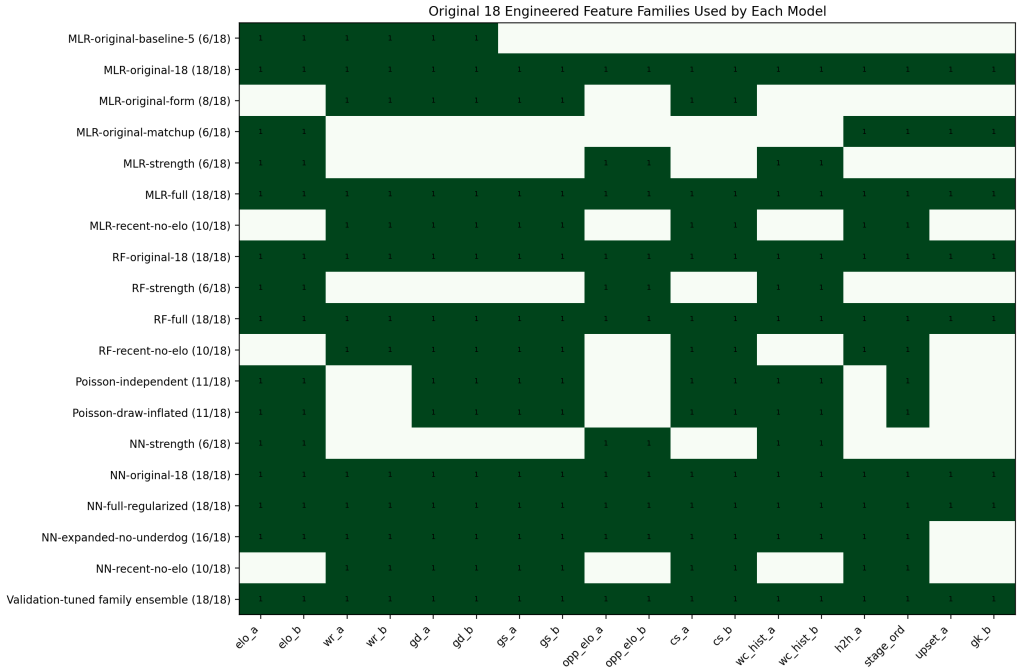

Figure 1: Feature-family usage by model. The original 18-feature models use all 18 directly; the expanded full MLR/RF/NN models use all 18 through 55 direct columns.

set becomes competitive once the training target includes all international matches. The best hard accuracy is still the NN-expanded-no-underdog ablation at 60.9%, with 0.9363 log loss and 30% draw recall.

Table 5: Final model comparison. CV summarizes 19 walk-forward World Cup checkpoints from 1938 through 2018; test is the held-out 2022 World Cup.

Model	Family	Feat.	Orig. 18	CV LL	CV Acc	Test LL	Test Acc	Draw Rec.
NN-original-18	NN	18	18	0.9657	0.5133	**0.9321**	0.5156	0.0000
NN-full-regularized	NN	55	18	0.9630	0.4832	0.9330	0.5312	0.0000
MLR-original-18	MLR	18	18	1.0183	0.4485	0.9340	0.4844	0.0000
Validation-tuned ensemble	Ensemble	70	18	**0.9360**	**0.5553**	0.9343	0.5469	0.0000
NN-strength	NN	8	6	0.9739	0.4566	0.9360	0.5000	0.0000
NN-expanded-no-underdog	NN	53	16	0.9646	0.5188	0.9363	**0.6094**	**0.3000**
NN-recent-no-elo	NN	28	10	0.9906	0.4712	0.9418	0.5000	0.0000
RF-original-18	RF	18	18	0.9991	0.4419	0.9475	0.5000	0.0000
Poisson-independent	Poisson	13	11	0.9526	0.5387	0.9479	0.5312	0.0000
MLR-full	MLR	55	18	1.0409	0.4496	0.9519	0.5312	0.1000
Poisson-draw-inflated	Poisson	13	11	0.9538	0.5367	0.9527	0.5312	0.0000
MLR-recent-no-elo	MLR	28	10	1.0337	0.4156	0.9546	0.4844	0.2000
MLR-original-matchup	MLR	6	6	1.0344	0.4261	0.9559	0.5312	0.0000
RF-full	RF	55	18	0.9926	0.4507	0.9562	0.5625	0.0000
RF-recent-no-elo	RF	28	10	1.0073	0.4475	0.9602	0.4844	0.1000
MLR-strength	MLR	8	6	1.0244	0.4297	0.9613	0.5312	0.1000
MLR-original-form	MLR	8	8	1.0193	0.4455	0.9637	0.4844	0.0000
MLR-original-baseline-5	MLR	5	6	1.0002	0.4442	0.9651	0.5469	0.1000
RF-strength	RF	8	6	1.0115	0.4027	0.9702	0.5312	0.0000

What succeeds and fails. The NN family succeeds because supervised international labels provide enough examples for nonlinear models. The 18-feature NN has the best 2022 log loss, while the 53-feature NN has the best 2022 accuracy. This is the central accuracy-vs-log-loss tradeoff introduced in Section 1: the best calibrated probability model is not the best hard classifier. Figure 3 shows why we report both metrics. Per-checkpoint log loss moves substantially across eras because tournament size, draw rates, and international data coverage change; a single average would hide

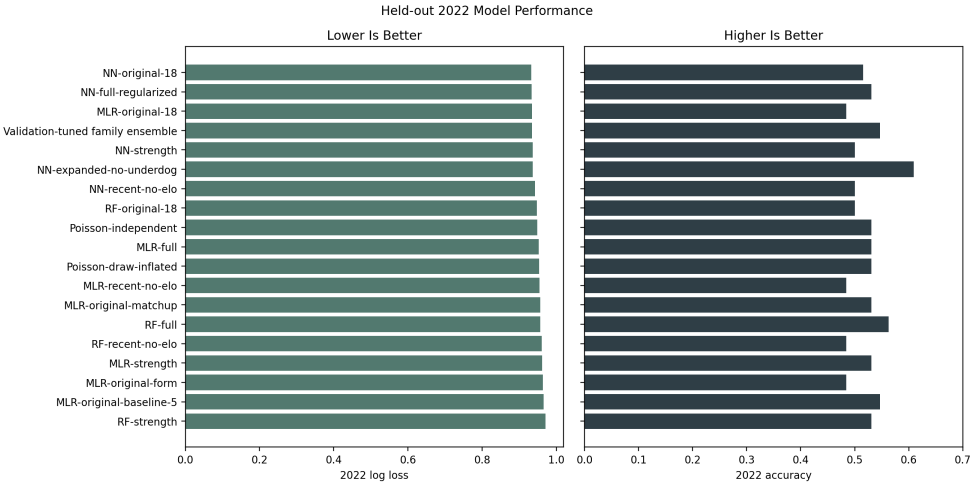

Figure 2: Held-out 2022 performance across trained models.

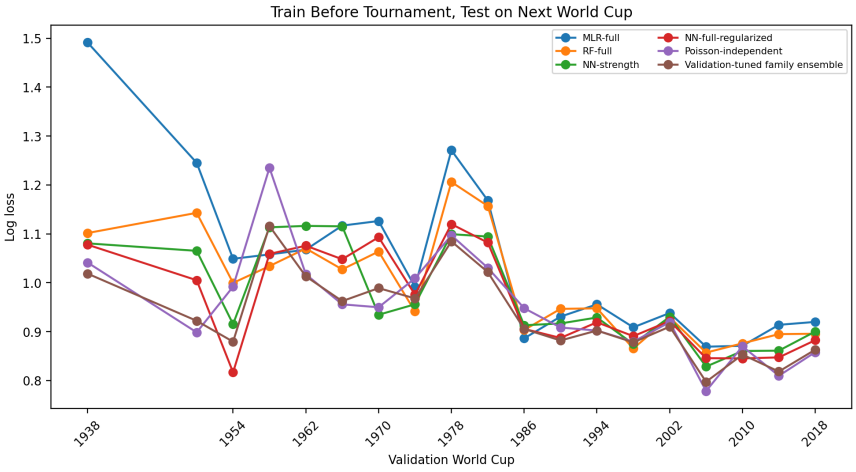

Figure 3: Walk-forward log loss by checkpoint. Each point trains before a World Cup and validates on that next World Cup.

this instability. MLR is stable but underfits nonlinear context, RF trails on log loss because tree probabilities are less smooth on small folds, and Poisson models are useful for simulation because they generate score distributions even when their argmax predictions rarely select draws.

6 2026 Forecast

The 2026 World Cup forecast uses a hybrid outcome/score simulator. The primary simulator uses NN-original-18, the best held-out log-loss model, for W/D/L probabilities. It separately fits a Poisson goal layer, then reweights the score grid so the total win/draw/loss mass matches the NN probabilities. This keeps the calibrated classifier in charge of match outcomes while still producing goals for group points, goal difference, goals for, and knockout advancement. Model selection uses only the historical validation checkpoints, but the final 2026 fit uses an augmented table of 19,239 rows: the original 14,881 rows plus 4,358 post-cache international matches from 2022 through March 31, 2026. These rows also update Elo and rolling-form state before the tournament simulation. The simulator reads the group field from `config/2026_groups.csv`, advances the top two teams in each group plus the eight best third-place teams, and then uses a transparent high-vs-low seeded knockout approximation. This approximation is reproducible but does not fully encode FIFA's exact round-of-32 pairing table.

Table 6: Representative 2022 confusion matrices. Rows are actual labels and columns are predicted labels.

Model	Actual	Pred W	Pred D	Pred L
NN-original-18	W	27	0	6
	D	7	0	3
	L	15	0	6
NN-expanded-no-underdog	W	30	0	3
	D	6	3	1
	L	15	0	6
Poisson-independent	W	32	0	1
	D	10	0	0
	L	18	0	3

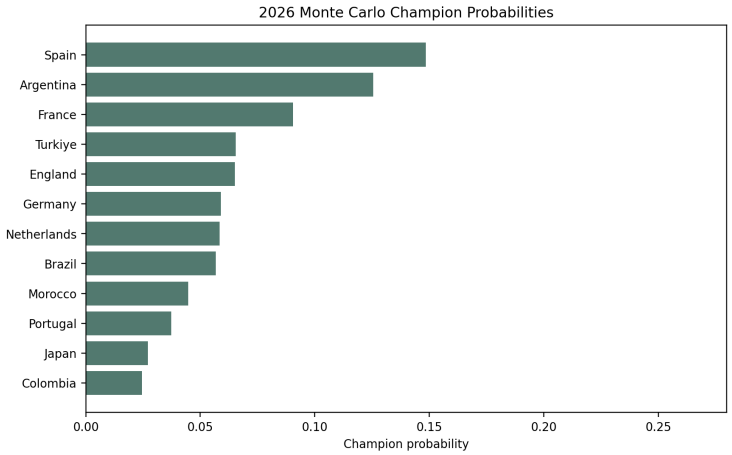

Figure 4: Top 2026 champion probabilities from 5,000 Monte Carlo simulations.

The top primary championship probabilities are Spain 14.9%, Argentina 12.6%, France 9.1%, Turkiye 6.5%, England 6.5%, Germany 5.9%, Netherlands 5.8%, Brazil 5.7%, Morocco 4.5%, and Portugal 3.7%. Polymarket placed Spain at 15.3% in the April 30, 2026 snapshot [Polymarket, 2026], so the primary model is close to the market's Spain estimate while assigning more probability to Argentina and less to France, England, Brazil, and Portugal. For sensitivity, the appendix reports all 48 champion probabilities for four forecast variants: Poisson-only, the primary NN-original-18 hybrid, the high-accuracy NN hybrid, and the validation-tuned ensemble hybrid. The spread across variants shows that winner forecasts are more model-sensitive than the 2022 W/D/L test metrics alone suggest.

The comparison with betting markets is useful because liquid odds aggregate information that is difficult for our feature table to encode, including squad depth, injuries, tactical news, public expert judgment, and path-specific expectations. In an efficient market, prices should approximate event probabilities after accounting for fees and overround, so a model that gives a top contender a probability far outside market consensus needs a technical explanation. Our primary Spain estimate being close to Polymarket is therefore a sanity check on calibration, not a proof that the model is correct. Markets can also fail: they can be distorted by liquidity, public-team bias, stale information, hedging behavior, and correlated trader beliefs. Our models differ because they are constrained to historical match results, engineered form features, Elo, and a simplified bracket. Where the model is above market, it is usually emphasizing recent performance and simulated path strength; where it is below market, it is usually missing roster quality, player availability, or market expectations about future team selection. The two views are therefore complementary: betting odds provide an external probability benchmark, while the model provides a reproducible explanation of which historical signals drive the forecast.

7 Related Work

Football score prediction often begins with Poisson goal models [Maher, 1982]. Dixon and Coles improve this family by correcting low-score dependence [Dixon and Coles, 1997]. Elo ratings are effective features for football outcome prediction [Hvattum and Arntzen, 2010]. Random forests and hybrid tournament models have also been applied to international football [Schauberger and Groll, 2018, Groll et al., 2019]. Tournament simulators commonly combine match probabilities with Monte Carlo brackets [Groll et al., 2015, FiveThirtyEight, 2022]. Our contribution is the course-project system design: all supervised international labels, walk-forward World Cup checkpoints, multiple model families, feature-family accounting, and a reproducible classifier-calibrated 2026 simulation.

8 Discussion

The project satisfies the main technical goals: it compares MLR, RF, NN, and Poisson variants; reports accuracy and log loss; uses all feasible World Cup checkpoints; and produces a full 2026 forecast. The strongest lesson is that model capacity only helps when the training target contains enough supervised labels. Here, the NN models are viable because they train on thousands of international matches while still validating on future World Cups. The broader implication is that sports prediction papers should avoid a single "best model" claim: calibrated probabilities and hard classification can favor different models, and tournament-era variability should be shown rather than averaged away.

The remaining limitations are concrete. First, the 2026-only supplement improves freshness but still lacks squad availability, injuries, roster depth, and market information. Second, the knockout bracket is approximated rather than exact. Third, the models still struggle with draws: the best accuracy NN predicts some 2022 draws, but most families do not. A production version should keep the 2022–2026 match feed current through the tournament, add official rankings and squad availability, calibrate probabilities explicitly, and encode the exact 2026 bracket. The submitted directory is self-contained, and the report stands alone as a complete description of the project.

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A Additional Figures and Tables

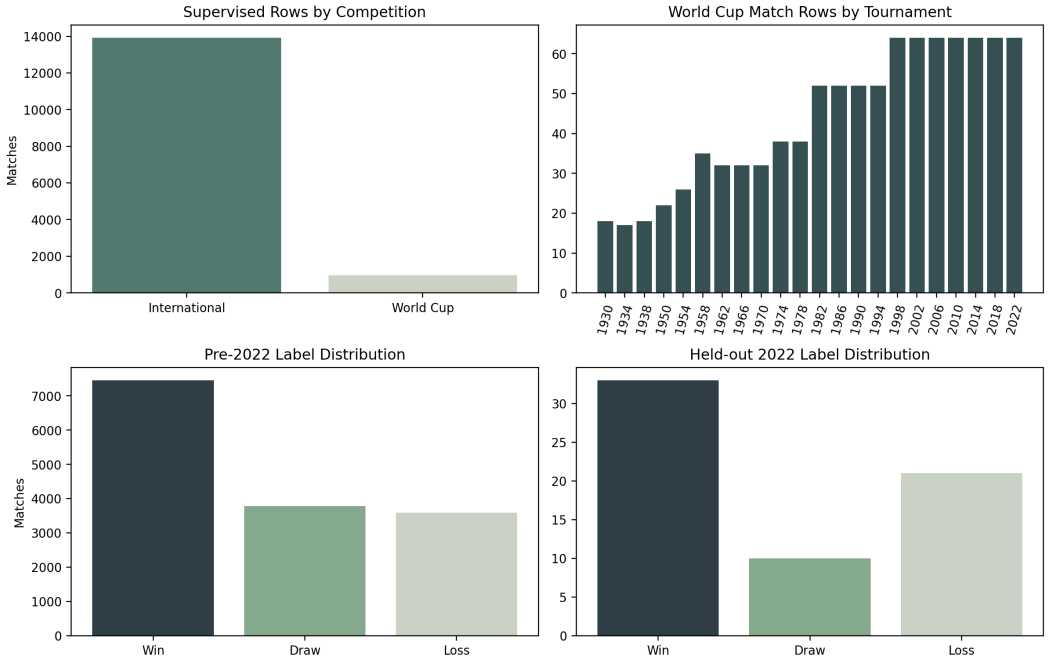

Figure 5: Dataset overview used by the final pipeline.

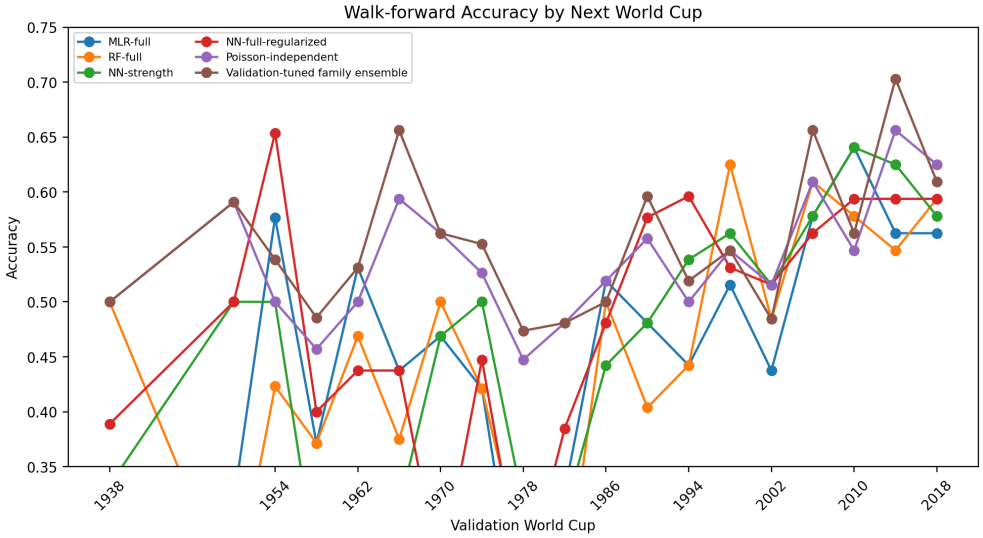

Figure 6: Walk-forward accuracy by validation World Cup.

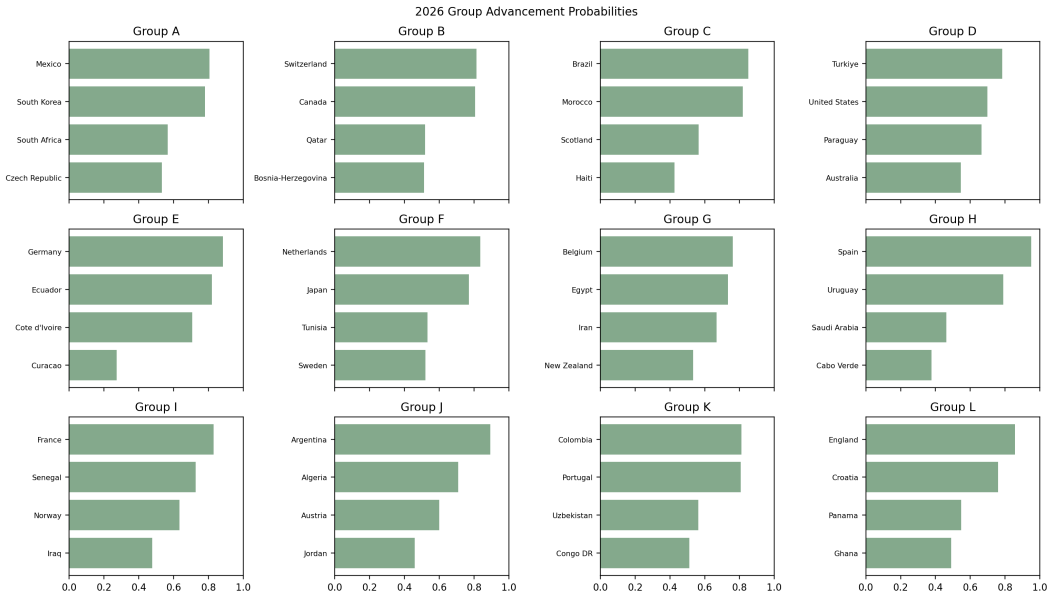

Figure 7: Simulated group advancement probabilities for the configured 2026 group field.

Table 7: 2026 champion probabilities for every forecast simulator, compared with Polymarket implied winner probabilities from the April 30, 2026 market snapshot [Polymarket, 2026]. Values are percentages. NN18 is the primary NN-original-18 calibrated score simulator; NN53 is the high-accuracy NN-expanded-no-underdog calibrated score simulator.

Team	Poisson	NN18	NN53	Ensemble	Polymarket
Spain	25.1	14.9	14.8	19.1	15.3
Argentina	19.7	12.6	9.0	15.0	8.7
France	13.0	9.1	6.7	9.6	16.4
Turkiye	2.6	6.5	2.7	3.5	0.7
England	6.9	6.5	7.0	6.6	11.1
Germany	4.1	5.9	4.7	4.9	5.2
Netherlands	4.3	5.8	4.1	5.4	3.4
Brazil	1.4	5.7	4.4	3.3	8.6
Morocco	4.8	4.5	5.5	4.0	1.9
Portugal	3.0	3.7	3.8	3.2	7.4
Japan	3.7	2.7	3.8	3.5	2.2
Colombia	1.0	2.5	3.3	2.0	1.7
Mexico	1.4	2.3	2.3	2.2	1.1
Belgium	0.8	1.6	1.0	1.5	2.0
Switzerland	0.5	1.6	1.9	1.1	1.0
Croatia	1.4	1.6	2.1	1.8	1.1
Uruguay	0.2	1.3	1.4	0.9	1.1
Ecuador	1.7	1.2	3.7	1.4	0.8
Senegal	0.6	1.0	1.6	1.3	0.8
Algeria	0.4	1.0	1.0	0.7	0.4
Paraguay	0.2	0.8	1.2	0.7	0.5
Iran	0.5	0.8	1.2	1.1	0.2
Canada	0.3	0.8	1.1	0.9	0.7
Norway	0.7	0.7	1.8	1.3	2.4
South Korea	0.4	0.7	1.3	0.9	0.5
United States	0.1	0.7	1.1	0.4	1.6
Egypt	0.1	0.6	0.7	0.4	0.5
Cote d'Ivoire	0.1	0.6	0.6	0.3	0.5
Australia	0.5	0.5	1.1	0.8	0.3
Sweden	0.0	0.3	0.5	0.1	0.7
Austria	0.2	0.3	0.5	0.6	0.7
Tunisia	0.0	0.2	0.4	0.1	0.4
Uzbekistan	0.2	0.2	0.6	0.5	0.3
Scotland	0.0	0.1	0.4	0.2	0.4
Panama	0.1	0.1	0.5	0.3	0.3
Haiti	0.0	0.1	0.1	0.1	0.2
Czech Republic	0.0	0.1	0.1	0.1	0.4
South Africa	0.0	0.1	0.1	0.0	0.3
Jordan	0.0	0.1	0.5	0.0	0.3
Congo DR	0.0	0.1	0.2	0.2	0.3
Iraq	0.0	0.1	0.3	0.1	0.3
Ghana	0.0	0.1	0.1	0.0	0.3
Bosnia-Herzegovina	0.0	0.0	0.1	0.0	0.4
Qatar	0.0	0.0	0.1	0.0	0.3
Cabo Verde	0.0	0.0	0.1	0.0	0.3
Curacao	0.0	0.0	0.0	0.0	0.3
New Zealand	0.0	0.0	0.4	0.1	0.3
Saudi Arabia	0.0	0.0	0.2	0.0	0.3

Table 8: Full glossary of the 18 engineered feature families.

Feature	Family	Definition
`elo_a`	Strength	Pre-match Elo rating of higher-rated team A.
`elo_b`	Strength	Pre-match Elo rating of lower-rated team B.
`wr_a`	Recent form	Team A win-rate over the recent-match form window.
`wr_b`	Recent form	Team B win-rate over the recent-match form window.
`gd_a`	Recent form	Team A average goal differential over the form window.
`gd_b`	Recent form	Team B average goal differential over the form window.
`gs_a`	Recent attack	Team A average goals scored over the form window.
`gs_b`	Recent attack	Team B average goals scored over the form window.
`opp_elo_a`	Schedule strength	Average recent opponent Elo faced by team A.
`opp_elo_b`	Schedule strength	Average recent opponent Elo faced by team B.
`cs_a`	Recent defense	Team A clean-sheet rate over the form window.
`cs_b`	Recent defense	Team B clean-sheet rate over the form window.
`wc_hist_a`	Tournament history	Time-decayed deepest prior World Cup stage reached by team A.
`wc_hist_b`	Tournament history	Time-decayed deepest prior World Cup stage reached by team B.
`h2h_a`	Matchup	Historical head-to-head score for team A against team B.
`stage_ord`	Context	Ordinal match-stage indicator from group stage through final.
`upset_a`	Upset tendency	Exponential moving-average rate at which team A lost matches while favored by Elo.
`gk_b`	Underdog tendency	Exponential moving-average rate at which team B won matches while unfavored by Elo.